Stage 3: Requirements Gathering

ML Diploma Graduation Project | Deliverable Template v1.0

Team name	Team 171
Project title	AI-Powered Predictive Maintenance for Industrial Equipment
Submission date	30 October 2026
Version	v1

1. Stakeholders and Users

Stakeholder	Relation to the system	What they need	How we gathered it
Maintenance engineers (intended users)	Inspect equipment condition and possible failure risk	A 30-cycle warning, interpretable risk information and sensor trends	Needs inferred from the project brief and Stage 1 user description; no interviews conducted
Maintenance planners (intended users)	Prioritize maintenance work	Clear machine/cycle identification and an actionable alert view	Needs inferred from the project overview and planned Streamlit dashboard
DEPI instructor / reviewer	Review project and milestone evidence	Measurable, traceable requirements and a reproducible demonstration	DEPI Stage 3 template and original project brief
Team 171 developers / model operator	Build, test and maintain the prototype	Engine-safe evaluation, comparison, logs, documentation and local deployment	Team proposal, Stage 1 objectives/WBS and Stage 2 literature review

2. Elicitation Summary

Sources and methods. Requirements were derived from the original DEPI predictive-maintenance brief, Team 171 proposal, Stage 1 objectives/scope/WBS, Stage 2 literature review and documented comparisons of Siemens, AWS and IBM tools. These are document analysis and secondary research, not first-hand interviews.

Key findings. A useful prototype needs a 30-cycle failure probability, leakage-safe FD001 preprocessing, comparable RF/XGBoost/LSTM results, a deployed API/dashboard, held-out sensor replay, MLflow tracking and basic monitoring. RUL estimation remains secondary.

Evidence limit. No maintenance-user interviews or surveys are documented. The stakeholder needs above are intended-user assumptions that should be confirmed with the instructor or representative users. The FD001 dataset has already undergone exploratory analysis and data-quality checks. Model performance, API latency, and deployment behavior will be verified during implementation and testing.

3. User Stories

ID	As a ..., I want ..., so that ...	Priority (MoSCoW)	Objective ID (S1)
US-01	As a maintenance engineer, I want a failure-within-30-cycles probability and warning so that I can prioritize inspection.	Must	O-01
US-02	As a maintenance planner, I want a dashboard showing engine ID, cycle, risk and sensor trends so that I can review engine condition.	Must	O-04
US-03	As an ML team member, I want RF, XGBoost and LSTM results on engine-separated data so that I can choose a defensible model.	Must	O-02
US-04	As a model operator, I want experiments, predictions and basic drift checks recorded so that results are reproducible and monitorable.	Must	O-05

ID	As a ..., I want ..., so that ...	Priority (MoSCoW)	Objective ID (S1)
US-05	As a DEPI reviewer, I want a cycle-by-cycle replay of a held-out engine so that I can verify that predictions update throughout the trajectory.	Must	O-04
US-06	As a maintenance planner, I want an estimated RUL when available so that I can also see approximate remaining cycles.	Could	O-03

4. Functional Requirements

ID	Requirement	Priority	Acceptance criterion
FR-01	The system shall import the official FD001 files and validate engine ID, cycle, 3 operating settings and 21 sensor columns.	Must	A schema check passes for supplied FD001 training and test data; missing required columns produce an explicit error.
FR-02	The system shall construct the 30-cycle training label and create engine-separated train/validation/test partitions without feature leakage.	Must	For full training trajectories, label = 1 when true RUL is 0–30 cycles; no engine overlaps across partitions; no future reading is in any input feature.
FR-03	The system shall prepare causal time-series features and LSTM windows, including rolling statistics and sensor-change trends.	Must	Feature tests on a sample engine show that changing future cycles cannot affect features for earlier cycles; PCA/frequency features are evaluated or their exclusion documented.
FR-04	The system shall train, tune and compare Random Forest, XGBoost and LSTM with engine-safe, time-series-aware validation.	Must	Saved evaluation compares all three on precision, failure-class recall/F1 and ROC-AUC; grouped/time-series validation and tuning avoid future leakage; an ensemble option is assessed or documented.
FR-05	The system shall provide a FastAPI prediction endpoint using the saved preprocessing and selected model to return 30-cycle failure risk.	Must	A valid request returns engine ID, cycle, probability in [0,1], predicted class and a validation-selected warning threshold; invalid inputs return a clear 4xx response.
FR-06	The system shall display engine condition, prediction and warning status in a Streamlit dashboard.	Must	Given a valid replay frame, dashboard shows engine ID, cycle, probability, risk status and recent sensor trends.
FR-07	The system shall replay a held-out FD001 engine cycle by cycle using only observations available up to each cycle.	Must	One complete test-engine replay runs through the API/dashboard in increasing cycle order without reading future cycles at inference.
FR-08	The system shall record training runs, parameters, metrics and model versions in MLflow.	Must	For each reported model, an MLflow run identifies the parameters, evaluation metrics, model artifact and data/split version.
FR-09	The system shall log predictions and run basic input-drift checks and provide documented retraining steps.	Must	Replay creates timestamped engine/cycle prediction logs; a documented drift check runs and yields a status; a retraining procedure is present in README.
FR-10	The system shall optionally output estimated remaining useful life (RUL) in operational cycles if the secondary model is completed.	Could	If enabled, response includes non-negative RUL estimate and a held-out MAE/RMSE report; if not built, it is clearly marked as not implemented.

5. Machine Learning Requirements

Prediction task (classification, regression, detection, ...)	Primary: binary classification (failure within the next 30 operational cycles). Secondary, optional: RUL regression if the core system remains on schedule.
Target or output, and its format	Binary target = 1 when remaining life is within the next 30 cycles, otherwise 0. Inference output: engine_id, cycle, predicted_failure_30_cycles (0/1), failure_probability (0–1), and warning status. Optional rul_cycles (non-negative number).
Inputs available at prediction time	Engine ID and current cycle; the 3 current operating settings and 21 current sensor values; available prior cycles of that same engine for rolling/sequence features. Exclude true RUL, failure labels, total eventual lifespan and any future readings.
Minimum performance (metric and target value)	From Stage 1 O-01, on held-out engines: failure-class recall ≥ 0.85, failure-class F1 ≥ 0.80, ROC-AUC ≥ 0.90. Report precision as well. Optional RUL O-03: RMSE ≤ 30 cycles (stretch MAE ≤ 25).
Baseline to beat	Use a simple logistic regression baseline (also compare majority-class prediction). Stage 1 O-02 requires the selected model to exceed the simplest baseline on validation F1; compare RF, XGBoost and LSTM under the same protocol.
Latency or throughput requirement	At least 95% of 50 API prediction requests should complete within 5 seconds after model warm-up in the local Docker environment.
Explainability requirement	Show failure probability and risk threshold; provide global feature importance/permutation importance for the selected model and document limitations. Do not claim a causal reason for failure.
Fairness or bias considerations	No demographic attributes are in FD001. Evaluate variability across held-out engines and operating-cycle ranges; state that simulated FD001 (one condition/fault mode) may not generalize to actual fleets or equipment types.
Retraining or update expectation	Stage 1 commits to MLflow versioning, drift checks and documented manual retraining. The original project brief additionally requires an automated retraining pipeline; this scope difference requires instructor confirmation before final design.

6. Data Requirements

Data item	Source	Volume	Update frequency	Quality or privacy constraint
FD001 training trajectories	NASA C-MAPSS FD001 public dataset	100 engines; 20,631 cycles; 21 sensors + 3 settings	Static download	Check engine/cycle ordering, missing/erroneous data and duplicates; do not fit preprocessing outside training folds.
FD001 official test + RUL ground truth	NASA C-MAPSS FD001 test and RUL files	100 truncated test trajectories + endpoint RUL labels	Static download	Hold out for final evaluation only; RUL truth may be used offline for scoring, never as prediction-time input.
Engineered features and training labels	Derived offline from FD001 training partitions	Up to 20,631 cycle-level examples, plus windows	Recomputed per pipeline run	Rolling features use past/current readings; labels may use future run-to-failure only during offline training.
Simulated live sensor frames	Held-out FD001 engine replay	One frame per recorded operational cycle	Each replay cycle	Only current/past sensor settings, engine ID and cycle are passed to inference; no live hardware.
Predictions and drift records	Local FastAPI/MLflow/monitor logs	Per inference or monitored replay batch	Per prediction / drift check	No personal information required; store engine ID, cycle, score, time and status only.

7. Non Functional Requirements

ID	Category	Requirement	How it will be measured
NFR-01	**Performance**	The application shall meet the proposed local inference latency target (95th percentile ≤ 5 s after warm-up).	Time 50 valid API calls on the documented Docker host; record p95 latency and hardware.
NFR-02	**Usability**	The dashboard shall visibly label the selected engine, cycle, 30-cycle probability and warning state.	Two team reviewers complete a documented demo checklist without asking what the displayed probability means.
NFR-03	**Security**	The prototype shall run locally by default without public exposure or hard-coded secrets; no production login is claimed.	Review Docker port binding and repository for secrets; confirm service is not publicly exposed in default setup.
NFR-04	**Privacy**	The system shall process public industrial simulation data and avoid logging personal identifiers.	Inspect 20 sample prediction logs for email, names or credentials; none should appear.
NFR-05	**Reliability**	The replay shall complete for a held-out engine; malformed API requests shall fail with actionable errors.	Run one full-engine replay without crash and five malformed-input tests that return explicit 4xx errors.
NFR-06	**Maintainability**	A clean clone shall provide reproducible installation, tests and model-serving instructions.	Reviewer uses README and documented Docker commands; unit/integration tests run, with model and preprocessing versions recorded.

8. Constraints and Assumptions

Schedule and resources. Follow the 15-week team plan and local Docker-first deployment; compute may be limited to available CPUs or shared/free notebook GPUs. Latency and other numeric nonfunctional thresholds introduced in Stage 3 are proposed acceptance targets, not measured results.
Data and evaluation. FD001 is simulated and static, not a genuine IoT stream. Keep engine IDs isolated between splits; fit normalizers/oversampling within training folds only; use official test ground truth only for final offline scoring. Check NASA data reuse terms before redistributing raw files.
Assumptions and approvals. A single engine provides ordered sensor observations during replay. The warning threshold will be chosen on validation data, not the test set. No direct user interviews are documented. The initial implementation will use simulated sensor replay and documented retraining procedures. Additional automation or IoT integration will depend on instructor feedback and available resources.

9. Out of Scope

Stage 1 exclusions. No FD002–FD004 training, physical IoT hardware/live production sensor feeds, cloud deployment, mobile app, user authentication or enterprise-scale asset-management integration in the first version.
Conditional work. Optional RUL regression is not guaranteed and will not delay the primary 30-cycle classifier.
Scope considerations. The first version focuses on a local deployment with simulated sensor replay, basic monitoring, and documented retraining. Automated retraining and physical IoT integration are outside the current proposed scope, subject to instructor approval.

10. Traceability Matrix

Requirement ID	Objective ID (S1)	Design component (S4)	Implemented in (S5)	Test case (S6)
FR-01	O-01 / O-02			
FR-02	O-01 / O-02			
FR-03	O-02			
FR-04	O-02			

Requirement ID	Objective ID (S1)	Design component (S4)	Implemented in (S5)	Test case (S6)
FR-05	O-01 / O-04			
FR-06	O-04			
FR-07	O-04			
FR-08	O-05			
FR-09	O-05			
FR-10	O-03			
NFR-01	O-04			
NFR-02	O-04			
NFR-03	O-04			
NFR-04	O-05			
NFR-05	O-04			
NFR-06	O-04 / O-05			